NETFLIX | Global Streaming Intelligence Suite – Device Usage Analytics

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The Device Usage Analytics Dashboard provides a comprehensive and interactive view of how users engage with a simulated global streaming platform across multiple device types. Built using Power BI and a synthetic dataset of approximately 2,394 records, the dashboard replicates real-world streaming analytics using a snowflake schema data model.

This dashboard focuses on analyzing how streaming behavior varies across devices such as mobile, smart TV, desktop, tablet, and gaming consoles. It helps answer key operational and behavioral questions, including:

- Which devices are most frequently used for streaming?
- How is total watch time distributed across devices?
- Which devices drive the highest engagement levels?
- How does completion behavior vary across device types?
- How does device usage differ across geographic regions?

Through KPI indicators, interactive slicers, and visual charts, the dashboard provides a clear and actionable understanding of device-level streaming behavior.

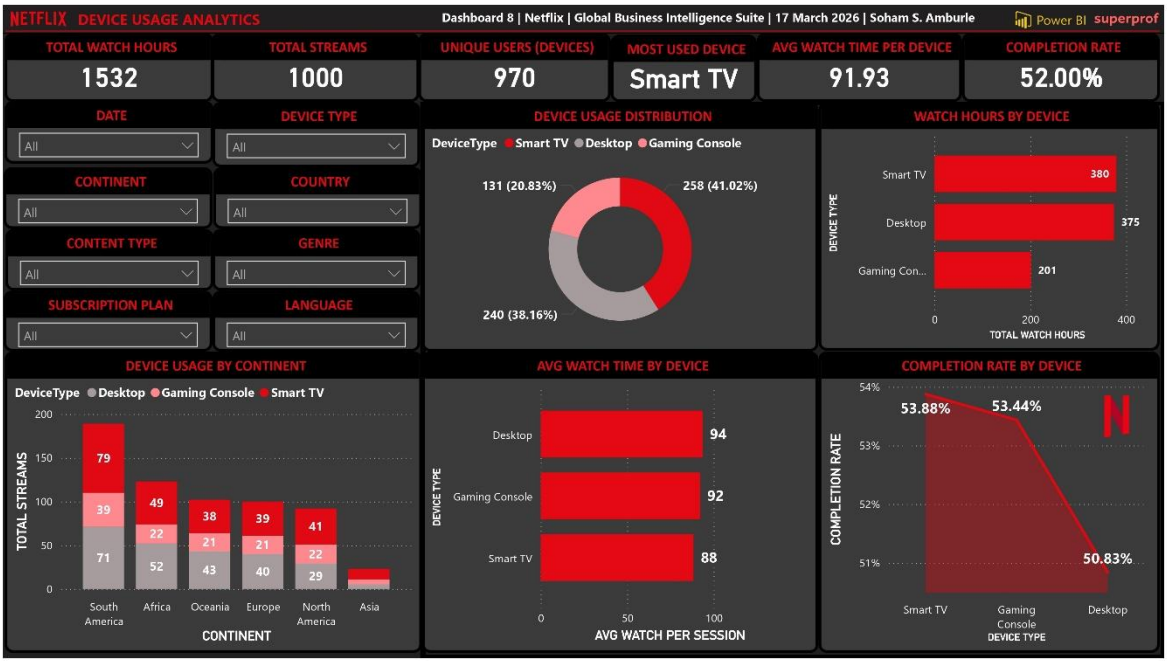

Fig 8.1 Device Usage Analytics

Dashboard Storyline

The dashboard follows a structured analytical flow, beginning with a high-level overview and progressing toward deeper insights into device-based usage patterns.

The first section presents key performance indicators that summarize overall platform usage across devices. These KPIs provide an immediate snapshot of total watch hours, streaming volume, user base, and engagement levels.

Data Disclaimer: This dataset is synthetic and created solely for learning, analysis, and portfolio demonstration. It does not represent real user data or actual activity from Netflix or any streaming platform.

The second section focuses on device distribution and engagement. Visualizations in this section help identify which devices dominate platform usage and how watch time is distributed across them.

The final section highlights regional device usage patterns and completion behavior. This allows stakeholders to understand how device preferences vary across continents and how engagement differs across platforms.

This structured approach ensures that users can quickly grasp the overall picture while also having the flexibility to explore deeper insights.

Dashboard Elements

The dashboard is composed of three primary analytical components:

- KPI Cards
- Interactive Slicers
- Visual Charts

These components work together to create a seamless and interactive analytical experience.

KPI Cards

Six KPI cards provide a summary of device usage metrics:

1. **Total Watch Hours**: Represents the total number of hours streamed across all devices.
2. **Total Streams**: Indicates the total number of streaming sessions.
3. **Unique Users**: Shows the number of distinct users engaging with the platform.
4. **Most Used Device**: Identifies the device with the highest usage.
5. **Average Watch Time per Device**: Displays the average watch duration per session.
6. **Completion Rate**: Represents the percentage of completed streaming sessions.

These KPIs provide a quick overview of platform engagement and device performance.

Slicers

Eight slicers enable dynamic filtering across multiple dimensions:

1. Date
2. Device Type
3. Continent
4. Country
5. Content Type
6. Genre
7. Subscription Plan
8. Language

These slicers allow users to analyze device usage across different regions, content types, and user segments, making the dashboard highly interactive.

Main Visualizations

The dashboard includes five key visualizations:

1. **Device Usage Distribution (Donut Chart)**: Shows the proportion of streaming sessions across different devices.

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2. **Watch Hours by Device (Bar Chart)**: Displays total watch time per device, highlighting engagement levels.
3. **Device Usage by Continent (Stacked Column Chart)**: Illustrates how device usage varies across geographic regions.
4. **Average Watch Time by Device (Bar Chart)**: Compares average session duration across devices.
5. **Completion Rate by Device (Area Chart)**: Highlights differences in completion behavior across devices.

Together, these visuals provide a clear and comprehensive view of device performance and user engagement.

Business Value

This dashboard delivers several analytical benefits:

- Identifies dominant streaming devices across the platform
- Evaluates engagement levels through watch time and session metrics
- Highlights completion trends to assess content consumption behavior
- Reveals regional variations in device usage
- Supports strategic decisions for platform optimization and device-specific improvements

Overall, the dashboard serves as a device intelligence layer within the broader streaming analytics ecosystem.

Tools and Data

- Visualization Tool: Power BI
- Dataset: Synthetic global streaming platform dataset
- Total Records: Approximately 2,394 rows across all tables
- Data Model: Snowflake schema
- Tables Used in the Dashboard:
    - Fact Tables
        - Fact_WatchHistory
        - Fact_Subscriptions
        - Fact_Ratings
    - Dimension Tables
        - Dim_Users
        - Dim_Content
        - Dim_Time
        - Dim_Geography
        - Dim_Genre
        - Dim_Language
        - Dim_Device
        - Dim_SubscriptionPlan

The dataset was generated using Mockaroo to simulate typical streaming analytics scenarios such as subscriber distribution, subscription plans, global geography, and viewing behavior.

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